Namaste – AI

pre-requisites

1) JS / Python
2) Basic node JS
3) CRUD AND API's
4) Basic GIT
5) React
6) Epress
7) Mongo

What is AI : making machines perform task that usualy requires human-Intelligence.

Machine Learning : Rules ✗
Examples ✓ [model learns on trainins data
Helps to make predictions and learn patterns.

DeepLearning : Neural Networks
Helps the computers thinks for themselves
Ex: Image Recognition
Translation
Speech Recognisatim
] More Practical Usage

GPU Revolution + Internet + Large datasets
↳ Better resources

ML : subset of AI that uses algorithm to analys data, learn and make decision.

1. Small datasets
2. structured data
3. Faster Training
4. Manual Feautre Enginery

vs

DL : subset of ML based on the Artificial Neural Networks

1. Vast datasets
2. Unstructured data
3. GPUs
4. Automated feature learn

Input data → Feautre Extractim → ML Algorithm (Decisimtree / Randm Forest) → Output

Ex: Spam detectim
House price predictim

Input data (unstructured) → Input layer → hidden layers → ouput layer → Output (Classification / predictn)

Ex: Image recognisation.
NLP (Translatim)

Computer vision Revolution. (dataset: Image Net)
↓
Trained large large Neutral Networks

Alexnet → deep Neutral Network (2012)

Next Ex: Face Unlock
X-Rays
Self driving cars
↳ see

Natural Language processing → we need machines to connect through words and understand.

0. Bag of words x (Good)
1. n-grams (Terribly-good) x
2. RNN (Recurrent Neutral Networks)
3. LSTM (Long-short term memory) → bettern than rnn.

Transformers (2017) **
paper → attention is all you need.

current stat:
Transformers — 2017
chatgpt — 2022
Agentic AI — 2026

LLMS:
Transformers trained on huge huge data.
GPU (Compute)
Data (Internet)

Generative AI:
classification
prediction
Recommendation
→ Generate (Multi-Model)
old vs New.

Chatgpt → LLMS (Nov 2022) **
(Open AI) + Transformers
→ Conversation
→ Made it Public
→ RLHF
→ Memory
→ The Whole AI movement

Multimodal AI:
Images
Voices
Videos
documents
Languages

Future of AI:
→ Agentic AI
→ Multimodel AI
→ Multi Agent Orchestration → Different Agents working together
→ Reasoning Models
→ RAG's
→ MCP / OKF
→ Robotics
→ Every possible thing/Industry

Todays AI:
Think and plan
write, Understand Code
Build and Test
Solve problems
work Autonomously

Ep: 3

Chatgpt knows it or guesses it?

Search Engines vs LLMs

Google → Retrieves

Question → search an index (→ faster) → rank relevant doc → result

LLMs → prompt → Learnt Patterns → Guessing/predicting next best → response.
↳ Generates

Search Engines → Retrieves (Existing pages)
LLMs → Generates (New content from already learnt patterns)

Search Engines

search → Index → Returns ranked documents

(Crawlers/spiders) → Internet

updated regularly.

Based on:
- domain authority
- page speed
- keywords
- avg time spent
- backlinks etc

→ SEO's → Always trail back to sources
↳ check who published it, at what time

LLMs:

↳ How LLMs generate responses:

- Predicts next word

Ex: The sun rises in [the east]
↳ generates

What knowledge does LLM contains:

→ A trained Neural Network contains a very large collection of numbers called parameters or weights

→ During training these numbers are repeatedly aligned.

→ Neural Networks are not trained everyday

Knowledge cutoff → Every LLM has its date.